

Music and Mental Health Analysis

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Stakeholder

The primary stakeholder for this memo is the Director of University Counseling & Wellness Services, who oversees mental-health outreach and student support strategy on campus. This stakeholder has some familiarity with data dashboards and survey-based evidence but does not work directly with statistical modeling, so they require clear interpretations rather than technical detail. They care about this analysis because understanding whether music-listening patterns are meaningfully related to depressive symptoms can help determine whether such questions should be incorporated into screening tools, campus wellness initiatives, or early-intervention outreach.

Executive Summary

This analysis examines how music-listening behaviors relate to depression symptoms using survey data from 718 respondents. The dataset was collected through an online Google Form distributed across social media platforms and public discussion forums. Respondents were predominantly young adults: the average age was approximately 24, and the median age was 21, indicating that the sample is largely representative of college-aged students.

We address two questions: (1) whether depression score (self reported on a scale of 0-10) varies based on hours of music listened to per day and streaming platforms, and (2) whether the number of hours spent listening to music each day is associated with the likelihood of reporting high depression symptoms (scores of 7 or above).

For the first question, we find that daily music-listening hours are related to depression scores, but the direction and strength of this relationship depend heavily on the streaming platform. In the baseline platform group (Apple Music), more listening time is associated with higher depression scores, while several other platforms show a weaker or even reversed pattern. This indicates that listening time alone is not a consistent or reliable indicator of depression. Other factors, such as insomnia symptoms, anxiety, and certain genre preferences, were more stable predictors of depression scores than listening hours. For the second question, our logistic regression model shows that daily listening hours do not meaningfully increase the odds of being in a high-depression category once other factors are taken into account. Instead, anxiety and insomnia serve as the strongest predictors of elevated depression symptoms.

Overall, although depression is associated with slightly higher listening time, daily music-listening hours are not a reliable indicator of high depression risk. Co-occurring mental-health symptoms provide far more informative predictive value than listening duration.

Problem Motivation

University counseling centers today face a clear and growing challenge: many students show signs of depression or anxiety, yet only a fraction seek help. A recent large study found that roughly 48.9% of college students in the post-pandemic era report depressive tendencies (Luo et al., 2024). Another analysis of more than 100,000 students across 64 studies estimated a pooled prevalence of depression symptoms at 33.6% and anxiety symptoms at 39.0%, demonstrating how widespread mental-health concerns are in higher education (Li et al., 2022).

Because counseling centers have limited resources, and because many students may not seek formal support, there is an urgent need for low-burden, behavior-based signals that could help identify or screen for potential mental-health risk. One promising candidate for such indicators is everyday music-listening behavior. Music listening is a near-universal daily behavior among young adults, and research suggests it may serve as a meaningful emotional regulator or mental-health resource. Studies on music's broader health and well-being effects have documented its benefits on mood, stress, immune function, and mental-health outcomes, including among individuals with diagnosed disorders (Rebecchini et al., 2021). More recent meta-analytic reviews find that informal "musicking", not only formal therapy, consistently supports emotion regulation in non-clinical populations (Peters et al., 2024), and music-based interventions have been shown to reduce symptoms of depression and anxiety (Rodwin et al., 2022).

Given these findings, everyday listening patterns may carry a measurable signal of underlying mental-health status, offering a low-cost way to complement traditional screening tools. If simple behaviors such as daily listening time, preferred streaming platforms, or music-use contexts reliably align with depression symptoms, counseling centers could incorporate brief questions about music habits into wellness check-ins or outreach initiatives. Such indicators would never replace clinical tools, but they could provide an accessible, early-warning layer that helps identify students who might otherwise go unnoticed.

Results and High-Level Findings

Overview of Key Findings

Across the full cleaned dataset (N = 646 Respondents for Model 1 and N = 718 Respondents for Model 2), several clear patterns emerged linking students' music-listening habits with their self-reported mental-health symptoms. While these relationships cannot establish causality, they reveal meaningful behavioral patterns that could inform low-burden wellness screening.

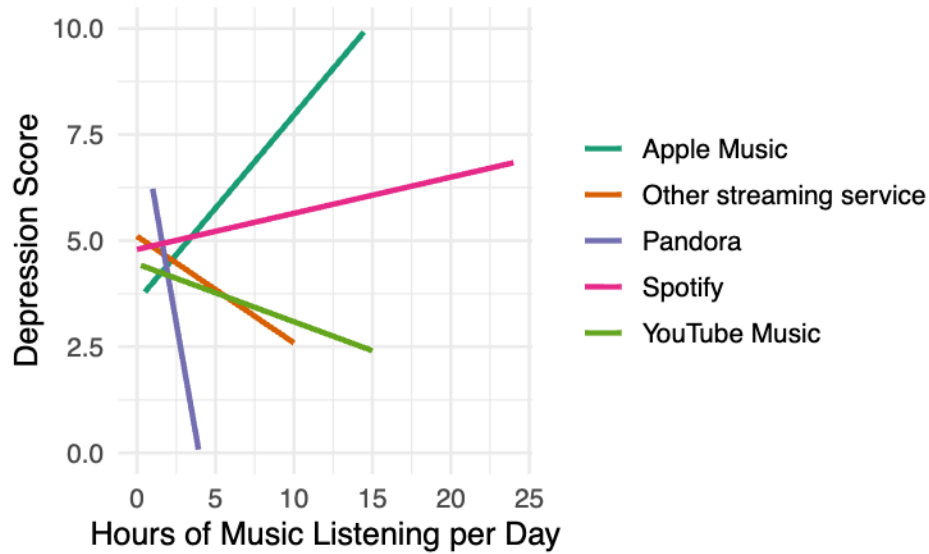


Figure 1: Daily Listening Time vs. Depression Score by Streaming Platform.

As shown in Figure 1, the relationship between daily music-listening time and depression score differs substantially across streaming platforms. For Apple Music users, the trend is strongly positive: students who listen for more hours per day show noticeably higher depression scores. Spotify users also show a positive association, though the slope is much more moderate. In contrast, students using YouTube Music or Other streaming services show a negative trend, where greater listening time is associated with lower depression scores for these groups. The pattern is most dramatic for Pandora, where the slope is sharply negative: even moderate increases in listening time correspond to substantially lower depression scores.

These contrasting slopes indicate that the effect of listening time on depression is highly platform-dependent, with some platforms showing the expected positive association and others showing the opposite. As a result, there is no single, uniform relationship between listening hours and depression symptoms across all listeners.

Drivers of Daily Listening Time

To examine predictors of depression scores, we fit a Poisson regression model. The final analytic sample included $N = 646$ respondents after excluding participants who did not report a primary streaming platform and removing two cases with non-integer depression scores (Poisson models require count outcomes).

The regression results show that daily listening time was a significant predictor of depression for users of the baseline streaming platform (Apple Music): each additional hour of music per day was associated with a 4.7% increase in expected depression scores. Independent of listening hours, several streaming platforms showed higher depression scores compared to the baseline service. Pandora users reported nearly 190% higher expected depression scores compared to the baseline platform, and Spotify users showed about a 24% higher depression score. Other streaming platforms and YouTube Music were not significantly different from the baseline at zero listening hours.

Mental-health symptoms were strong predictors of depression. Anxiety was the most influential: for each one-unit increase in anxiety, expected depression scores increased by 11%. Insomnia also had a meaningful effect, with each unit increase associated with a 4.9% rise in depression. OCD symptoms, by contrast, were

not a significant predictor after adjusting for the other variables. Most genre preferences were not associated with depression, but gospel listeners had strikingly lower scores: their expected depression was about 49% lower than listeners of the baseline genre, Classical.

Indicators of High Depression Symptoms

To assess whether music behavior can meaningfully signal elevated depression, we fit a logistic regression model predicting whether a respondent scored 7 or above on the 0–10 depression scale using 718 students. The updated results indicate that daily listening hours are still not a statistically significant predictor of high depression. The odds ratio for listening time is 1.06, meaning that each additional hour of listening is associated with a 5.8% increase in the odds of high depression, but this effect is not significant ($p = 0.072$). Age, musician status (instrumentalist or composer), and most genre preferences also showed no meaningful association with the likelihood of high depression.

In contrast, anxiety and insomnia remain the strongest predictors of elevated depression symptoms. For anxiety, the odds ratio is 1.39, meaning that each one-point increase raises the odds of high depression by 39% ($p < 0.001$). Insomnia also shows a robust effect: each unit increase is associated with a 15% increase in the odds of high depression ($p < 0.001$). Two additional predictors reached significance. Students who primarily listen to hip hop had more than three times the odds of high depression, whereas those who reported that music worsens their mood had odds more than four times higher than students who reported positive or neutral effects. OCD symptoms showed a marginally negative trend but did not reach significance.

Taken together, these results show that music-listening hours do not reliably distinguish students with high depression symptoms, even under a stricter severity threshold. Instead, validated psychological indicators—especially anxiety and insomnia—remain far more informative. Music behaviors, including genre preferences and streaming hours, may provide supplemental context but should not be interpreted as standalone mental-health signals.

Limitations and Recommended Next Steps

Although music-listening behaviors do contain some signal, our results show that the relationship between depression scores and daily listening hours is not consistent across streaming platforms. For some platforms, such as the baseline service and Spotify, more listening hours are associated with higher depression scores, while for others—such as YouTube Music, Pandora, and “Other” platforms—the trend is weaker or even reversed. This platform-dependent pattern means that music behavior alone is not a reliable standalone indicator of depression, because the meaning of “listening more” fundamentally changes depending on where students listen. Any practical use of these insights would need to be paired with validated psychological information rather than used in isolation.

Two substantive limitations of our data should be kept in mind when interpreting these findings. First, all mental-health measures rely on single self-reported items rather than standardized clinical scales like the PHQ-9 or GAD-7. These coarse, one-item scores may underestimate or overestimate true symptom severity, limiting how confidently we can draw inferences about the size or importance of predictors. Second, the dataset is cross-sectional, capturing students’ listening habits and mental-health symptoms at a single point in time. This prevents us from determining whether heavier or lighter listening patterns reflect coping strategies, consequences of mood changes, or stable personal preferences unrelated to mental-health trajectories.

Despite these limitations, the patterns suggest meaningful directions for future wellness monitoring and outreach. Because anxiety and insomnia—validated co-indicators of depression—were far more consistent predictors of depression scores than listening behavior, counseling services may benefit from prioritizing these co-occurring symptoms in early screening or brief questionnaires. Music-related responses may still play a useful supporting role, especially if collected with more nuance than a single “hours per day” measure. Our data do not capture why, when, or how students use music, and these contextual dimensions may contain information that simple duration-based measures miss.

Moving forward, the counseling center could incorporate a few targeted questions about students’ music use into existing wellness screenings—for example, whether they listen to manage stress, regulate mood, support sleep, or reduce intrusive thoughts. These specific motivations are more likely to relate meaningfully to mental-health needs than overall listening volume. Finally, the university could consider collecting higher-quality, campus-specific data using improved measures and representative sampling. Doing so would allow staff to evaluate whether certain music-use patterns provide early indicators of mental-health challenges within their own student population and develop more evidence-informed outreach strategies.

Technical Appendix

Data Background

The data used in this analysis come from the Music and Mental Health (MxMH) Survey Results hosted on Kaggle. The dataset was collected through an online Google Form distributed through social media platforms and public forums, resulting in a convenience sample of 736 respondents. After cleaning, participants ranged in age from 10 to 73, with an average age of approximately 24 and a median age of 21, indicating that the sample is largely composed of late-adolescent and young adult respondents. Because most individuals fall within the traditional college-age range, it is reasonably aligned with the population of interest for our stakeholder—a university counseling center—though it is not a representative campus sample. The dataset includes demographic variables (such as age), music-listening behaviors (hours per day, primary streaming platform, and preferred genres), and self-reported mental-health symptom scores for anxiety, depression, insomnia, and OCD (each rated 0–10). As a cross-sectional, self-reported dataset, it is appropriate for identifying associations but cannot establish causal relationships.

Data Cleaning

We began by assessing missingness across all variables. Overall, 107 observations had missing or inconsistent values in the BPM variable, and because responses to this item were highly unreliable, we removed the BPM column entirely. We additionally removed 1 observation with a missing age value to maintain completeness of demographic information. During outlier screening, we excluded 1 respondent whose reported age and music-listening hours were implausible and would have exerted undue influence on later models.

Several variables with binary “Yes/No” responses contained invalid or misspelled entries; to ensure consistency in these predictors, we removed 9 rows that did not provide valid responses. The music-effects variable contained 7 blank entries, which were also removed because the variable is used in downstream modeling. For analyses involving streaming platforms (RQ1), we excluded respondents who reported not using a streaming service or left the platform question blank, resulting in the removal of 60 observations for that subset. In addition, because Poisson modeling was used for RQ1, we excluded 2 rows that contained non-integer values for Depression score.

To prepare the binary outcome for RQ2, we created a “high depression” indicator classifying respondents scoring 7 or higher on the 0–10 depression scale as experiencing elevated symptoms. Because there is no established clinical cutoff for single-item self-reported depression scores, we adopted a data-driven threshold based on the sample’s third quartile ($Q3 = 7$), which identifies respondents in the highest 25% of the depression distribution.

After all cleaning steps, the final dataset contained 646 respondents for the model examining depression scores (RQ1) and 718 respondents for the model predicting high-depression status (RQ2).

Models

Research Question 1: “How does depression score vary by hours spent listening to music per day and primary streaming service?”

To examine how depression scores vary based on daily music-listening hours and streaming platform, we fit a Poisson regression model using the depression score (0–10) as the count-based response variable. Poisson

regression is appropriate because the outcome is a non-negative count, and the goal is to estimate how the expected depression score changes with listening time and platform use while adjusting for relevant covariates. We included an interaction between hours per day and primary streaming service to test whether the association between listening duration and depression differs across platforms. This directly addresses our research question by modeling whether the slope relating listening time to depression score varies depending on the service a respondent uses.

Beyond the primary predictors of listening hours and streaming platform, we incorporated a set of covariates to reduce potential confounding and better isolate the effects of platform-specific listening patterns. These covariates were chosen because they relate to both music-use behavior and broader mental-health characteristics. The model adjusts for age, whether the respondent listens while working, instrumentalist status, composer status, exploratory listening tendencies, listening to music in foreign languages, and three additional mental-health symptom scores (anxiety, insomnia, and OCD). We also included favorite genre and perceived music effects to account for musical preferences and how individuals believe music influences their emotions. Including these variables helps adjust for individual differences in musical habits, mental-health co-morbidity, and listening contexts that could bias estimated associations.

The Poisson framework assumes that the mean and variance of the outcome are equal, observations are independent, predictors have a log-linear relationship with the expected count, and predictors are not highly collinear. We evaluated these assumptions through diagnostic checks, including residual plots, variance-inflation factors, and an AIC comparison with a negative binomial model. Although the model showed mild overdispersion, the Poisson specification still provided a reasonable structure for examining platform-specific differences in the relationship between listening time and depression.

Research Question 2: “Does the number of hours spent listening to music each day increase the likelihood of reporting high depression symptoms?”

For our second research question, which sought to determine if the number of hours spent listening to music each day increases the likelihood of reporting high depression symptoms, we fit a logistic regression model with a binary outcome. Logistic regression is appropriate because the response variable, high depression, classifies respondents as either above or below a threshold level of self-reported depressive symptoms, and the model estimates how the log-odds of being in the high-depression category change with daily listening time.

Outside of our outcome variable of high depression and our primary independent variable, daily hours spent listening to music, we also included several covariates. Specifically, we adjusted for age, indicators of music engagement (instrumentalist and composer), co-occurring mental-health conditions (anxiety, insomnia, and OCD), and two global music-use characteristics (favorite genre and music effects). These variables may influence both how much an individual listens to music and their underlying likelihood of experiencing elevated depressive symptoms, thereby helping reduce confounding effects.

Logistic regression, and therefore our model, assumes independent observations, linearity of continuous predictors on the log-odds scale, and the absence of multicollinearity, all of which were checked via plots and tests.

Assessment and changes

Research Question 1

The first model we fit for RQ1 was a linear regression model that regressed depression score on our chosen predictors, including our interaction term of hours spent listening per day x primary streaming service.

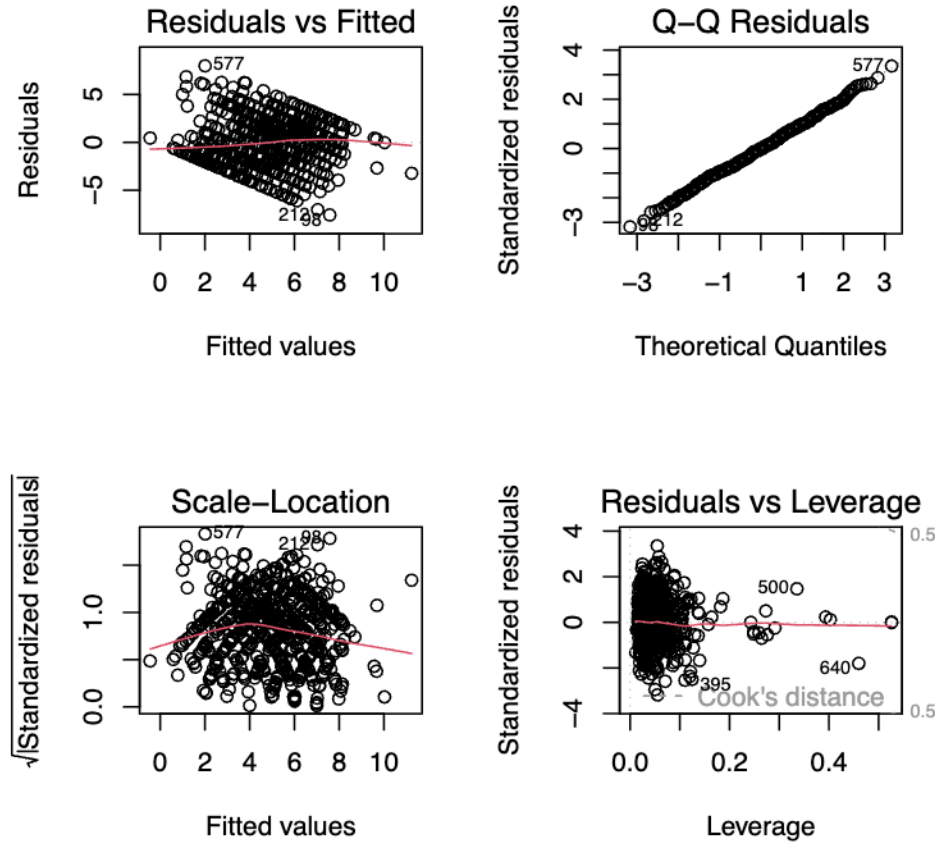


Figure 2: Diagnostic plots for the raw RQ1 model.

After fitting this initial model, we first examined the standard diagnostic plots, as shown in Figure 2: Residuals vs Fitted, Normal Q-Q, Scale-Location, and Residuals vs. Leverage. While the Normal Q-Q plot showed no strong evidence of a normality violation, as there are no significant departures from the 45 degree line, the Residuals vs Fitted plot revealed a strong violation of homoscedasticity, with the points forming a funnel-shape, and a small violation of linearity, where the red smoothing line curved downwards rather than remaining flat.

Because the linear regression model exhibited violations of both linearity and homoscedasticity, we decided to fit a Poisson model instead. After removing the two observations with non-integer depression scores, which are incompatible with count-based modeling, the dataset for RQ1 included $N = 646$ respondents.

Model	Df	AIC
Poisson	36	3112.210
Negative Binomial	37	3097.819

Table 1: AIC comparison between the Poisson and Negative Binomial models for RQ1

After fitting the Poisson model, we next evaluated whether its equidispersion assumption was reasonable. Because Poisson regression requires the mean and variance of the outcome to be equal, violations of this assumption can lead to underestimated standard errors and misleading significance tests. To assess this, we fit a corresponding Negative Binomial model, which allows the variance to exceed the mean, and compared the two models using AIC to determine whether overdispersion was present. As shown in Table 1, the AIC value of the poisson model is only slightly higher than that of the negative binomial model (3112.2 vs 3097.8), indicating some overdispersion but not a dramatic violation. Given the modest difference, we retained the Poisson model as our primary specification but note that the Negative Binomial model provides a slightly better fit.

Next, we assessed multicollinearity using GVIF values, which adjust for the degrees of freedom of multi-level categorical variables. No predictors exhibited adjusted GVIF values at levels that would indicate problematic multicollinearity, as they were all under 3.

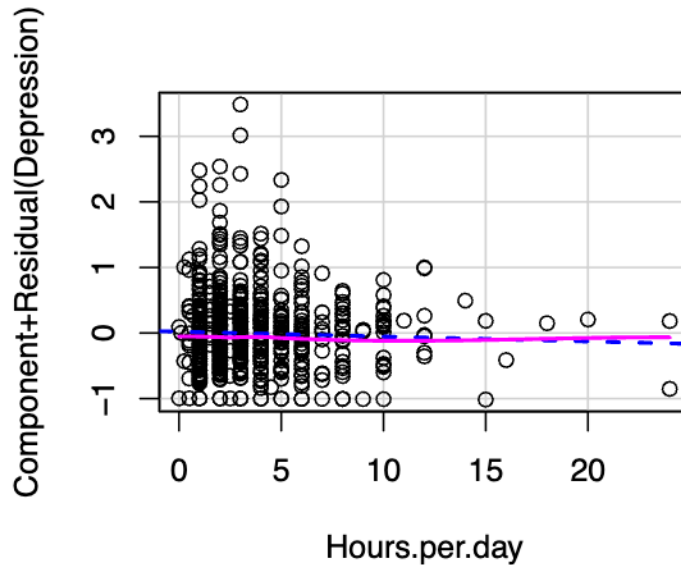


Figure 3: Partial residual plot for Hours.per.day.

To assess whether Hours.per.day was reasonably modeled as a linear predictor on the log scale, we examined a component-plus-residual (partial residual) plot from the Poisson model without the interaction term (Figure 3). Partial residual plots help evaluate whether a predictor’s effect follows a linear pattern, as assumed in Poisson regression. In our plot, the nonparametric smoother (pink line) aligns closely with the fitted linear component (blue dashed line) across the majority of the observed range. This indicates that the linearity assumption for listening hours is largely acceptable. The smoother shows only slight downward curvature

at higher listening times—where data are sparse—which suggests minor deviation from perfect linearity but not enough to indicate a substantial misspecification. Overall, the partial residual plot supports the use of a linear term for Hours.per.day in the Poisson model, with only mild nonlinear behavior at extreme values.

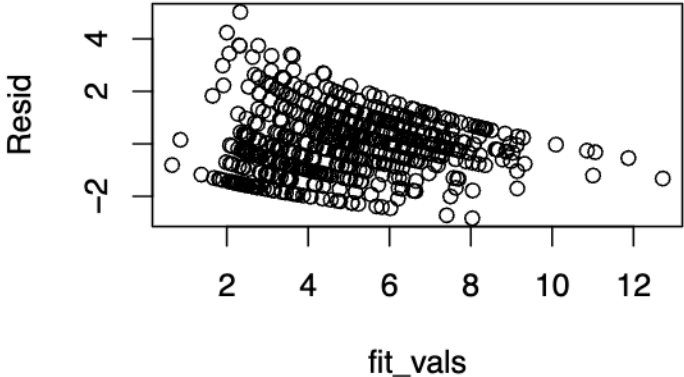


Figure 4: Pearson residuals vs. fitted values.

Finally, we examined the Pearson residuals plotted against the fitted values to evaluate overall model fit. In a well-specified Poisson regression, the residuals should appear randomly scattered around zero. Instead, as shown in Figure 4, the residuals form a clear pattern, with a sloping band of points rather than random noise. This indicates that important relationships in the data are not fully captured by the model. While the Poisson regression provides a reasonable approximation for our purposes, the structure in the residuals highlights that the true relationship between predictors and depression scores is more complex than the model assumes.

Term	Estimate	Std. Error	p-value	Exp(Estimate)	2.5%	97.5%
\(Intercept\)	0.4761	0.1555	0.0022	1.6097	1.1841	2.1782
Hours per day	0.0456	0.0161	0.0047	1.0467	1.0129	1.0792
Primary streaming service: Other streaming service	0.2652	0.1547	0.0864	1.3037	0.9612	1.7631
Primary streaming service: Pandora	1.0675	0.4089	0.0090	2.9081	1.2998	6.4954
Primary streaming service: Spotify	0.2164	0.1022	0.0342	1.2416	1.0183	1.5202
Primary streaming service: YouTube Music	0.1604	0.1281	0.2104	1.1740	0.9139	1.5104
Age	0.0002	0.0021	0.9051	1.0002	0.9962	1.0043
While workingYes	0.0498	0.0495	0.3142	1.0511	0.9546	1.1589
InstrumentalistYes	0.0015	0.0459	0.9741	1.0015	0.9149	1.0954
ComposerYes	0.0451	0.0544	0.4067	1.0461	0.9398	1.1631
ExploratoryYes	-0.0005	0.0451	0.9919	0.9995	0.9155	1.0925
Foreign languagesYes	0.0351	0.0390	0.3682	1.0357	0.9597	1.1180
Anxiety	0.1045	0.0080	0.0000	1.1101	1.0930	1.1277
Insomnia	0.0475	0.0063	0.0000	1.0487	1.0358	1.0617
OCD	-0.0081	0.0066	0.2234	0.9920	0.9791	1.0049
Favorite genre: Country	-0.0033	0.1299	0.9799	0.9967	0.7700	1.2823
Favorite genre: EDM	0.0946	0.1111	0.3945	1.0993	0.8838	1.3667
Favorite genre: Folk	0.0154	0.1156	0.8940	1.0155	0.8089	1.2729
Favorite genre: Gospel	-0.6753	0.2916	0.0206	0.5090	0.2777	0.8760
Favorite genre: Hip hop	0.1917	0.1082	0.0763	1.2113	0.9801	1.4980

Favorite genre: Jazz	-0.0749	0.1327	0.5723	0.9278	0.7126	1.1997
Favorite genre: K pop	-0.2591	0.1374	0.0592	0.7717	0.5870	1.0065
Favorite genre: Latin	-0.0464	0.3464	0.8935	0.9547	0.4494	1.7756
Favorite genre: Lofi	0.2126	0.1488	0.1531	1.2368	0.9183	1.6469
Favorite genre: Metal	0.0167	0.0933	0.8575	1.0169	0.8484	1.2233
Favorite genre: Pop	-0.0414	0.0912	0.6501	0.9595	0.8040	1.1497
Favorite genre: R and B	-0.1268	0.1200	0.2905	0.8809	0.6951	1.1130
Favorite genre: Rap	0.0473	0.1362	0.7282	1.0485	0.7999	1.3650
Favorite genre: Rock	0.0588	0.0855	0.4919	1.0605	0.8991	1.2574
Favorite genre: Video game music	-0.1071	0.1174	0.3616	0.8984	0.7129	1.1298
Music effects: No effect	0.0414	0.0465	0.3732	1.0423	0.9508	1.1411
Music effects: Worsen	0.3753	0.0991	0.0002	1.4555	1.1926	1.7591
Hours per day:Primary streaming service: Other streaming service	-0.1043	0.0396	0.0085	0.9010	0.8328	0.9729
Hours per day:Primary streaming service: Pandora	-0.5566	0.2240	0.0130	0.5732	0.3563	0.8628
Hours per day:Primary streaming service: Spotify	-0.0484	0.0174	0.0054	0.9527	0.9215	0.9867
Hours per day:Primary streaming service: YouTube Music	-0.0817	0.0272	0.0026	0.9215	0.8732	0.9715

Table 2: Final Poisson model statistics for RQ1.

Research Question 2

For RQ2, we modeled a binary outcome indicating “high depression” (depression score = 7) using logistic regression, with daily listening time (Hours per day) as our primary predictor.

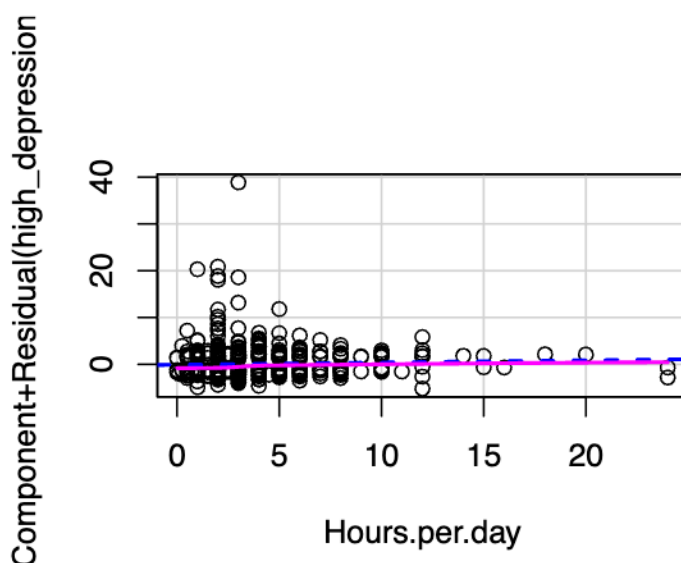


Figure 5: Partial Residuals Plot for Hours.per.day.

Because logistic regression assumes that continuous predictors have a linear relationship with the log-odds of the outcome, we began by checking this assumption for Hours per day using a component-plus-residual (partial residual) plot (Figure 5). This plot did not reveal strong curvature or systematic departures from linearity, as the pink line was fitted very tightly to the blue, so we retained Hours per day in linear form.

We then examined several additional diagnostics. First, we used VIFs to assess multicollinearity among the predictors and did not detect problematic levels of collinearity. Second, we evaluated overall model fit with

a Hosmer–Lemeshow goodness-of-fit test, which revealed a p-value greater than 0.05 (0.6483), meaning the test indicates no substantial issue with the fitting of the model.

We calculated leverage values using the hat matrix and identified several observations above the common threshold of $2\bar{h}$. However, high leverage alone is not sufficient evidence to remove points in logistic regression, particularly in models with many categorical predictors. Because these cases did not show extreme Pearson residuals either, we retained all observations for the final model.

TN	FP	FN	TP	Accuracy	Kappa	Sensitivity	Specificity	Precision	Recall	F1	Balanced Accuracy
396	120	71	131	0.734	0.387	0.522	0.848	0.649	0.522	0.578	0.685

Table 3: Confusion Matrix and Classification Metrics, Threshold = 0.5.

To assess predictive performance, we computed a confusion matrix (Table 3) using the conventional 0.5 probability threshold. The model produced 396 true negatives, 131 true positives, 71 false negatives, and 120 false positives, yielding an overall accuracy of 0.734. Sensitivity was 0.522, meaning the model correctly identified about 52% of respondents with high depression, while specificity was 0.848, indicating a much stronger ability to correctly classify low-depression respondents. Precision was 0.649, and the resulting F1 score of 0.578 reflects moderate performance with notable trade-offs between identifying true positives and avoiding false alarms. Overall, the model performs reasonably well at ruling out low-depression cases but struggles more with detecting high-depression respondents, as shown by the lower sensitivity and higher false-negative rate.

To further assess model performance, we examined the ROC curve and computed the AUC. The model achieved an AUC of 0.784, indicating strong discriminative ability. The optimal classification threshold was identified as 0.326, which balances sensitivity and specificity more effectively than the default 0.50 cutoff.

TN	FP	FN	TP	Accuracy	Kappa	Sensitivity	Specificity	Precision	Recall	F1	Balanced Accuracy
303	52	164	199	0.699	0.4	0.793	0.649	0.548	0.793	0.648	0.721

Table 4: Confusion Matrix and Classification Metrics (Threshold = 0.456).

At the optimized probability threshold determined from the ROC curve, we recomputed the confusion matrix (Table 4). Sensitivity increased substantially to 0.793, meaning the model correctly identified nearly 80% of respondents with high depression. Specificity decreased to 0.649, reflecting a higher rate of false positives but still acceptable performance given the shift toward prioritizing detection. Overall accuracy was 0.699, and the F1 score improved to 0.648, indicating a better balance between precision and recall at this threshold. This re-calibration increased the model’s ability to detect high-depression cases—an important improvement for applications where missing true cases is more costly than flagging additional false positives.

Term	Odds Ratio	Std Error	statistic	p-value	2.5% CI	97.5% CI
\(Intercept\)	0.03	0.56	-6.0173084	p<0.001	0.01	1.000000e-01
Hours per day	1.06	0.03	1.7977607	0.072	1.00	1.130000e+00
Age	0.99	0.01	-1.1676943	0.243	0.97	1.010000e+00
Instrumentalist: Yes	1.09	0.22	0.3866914	0.699	0.70	1.680000e+00
Composer: Yes	1.13	0.26	0.4833434	0.629	0.68	1.890000e+00
Anxiety	1.39	0.04	7.9922695	p<0.001	1.29	1.510000e+00
Insomnia	1.15	0.03	4.4168354	p<0.001	1.08	1.220000e+00

OCD	0.94	0.03	-1.7805888	0.075	0.88	1.010000e+00
Favorite genre: Country	1.07	0.61	0.1131878	0.910	0.31	3.520000e+00
Favorite genre: EDM	1.82	0.54	1.1022006	0.270	0.63	5.350000e+00
Favorite genre: Folk	1.37	0.55	0.5801116	0.562	0.47	4.040000e+00
Favorite genre: Gospel	0.45	1.30	-0.6128691	0.540	0.02	4.520000e+00
Favorite genre: Hip hop	3.40	0.54	2.2756192	0.023	1.20	9.990000e+00
Favorite genre: Jazz	1.30	0.62	0.4221057	0.673	0.38	4.420000e+00
Favorite genre: K pop	0.77	0.63	-0.4101026	0.682	0.21	2.620000e+00
Favorite genre: Latin	0.00	584.50	-0.0240369	0.981	NA	4.617628e+33
Favorite genre: Lofi	3.19	0.79	1.4784549	0.139	0.70	1.607000e+01
Favorite genre: Metal	1.27	0.44	0.5363145	0.592	0.54	3.040000e+00
Favorite genre: Pop	0.77	0.44	-0.5880957	0.556	0.33	1.850000e+00
Favorite genre: R and B	0.68	0.61	-0.6280024	0.530	0.20	2.220000e+00
Favorite genre: Rap	0.88	0.67	-0.1905819	0.849	0.22	3.170000e+00
Favorite genre: Rock	1.54	0.40	1.0756764	0.282	0.71	3.460000e+00
Favorite genre: Video game music	0.47	0.55	-1.3788966	0.168	0.15	1.360000e+00
Music effects: No effect	1.38	0.22	1.4531248	0.146	0.89	2.140000e+00
Music effects: Worsen	4.53	0.57	2.6491119	0.008	1.52	1.471000e+01

Table 5: Logistic regression model for RQ2 (odds ratios with 95

Overall, the model shows that anxiety and insomnia are the strongest and most consistent predictors of high depression symptoms, with both variables having statistically significant odds ratios above 1. In contrast, hours of music listening, demographic factors (including age), musician status, and most favorite-genre categories do not demonstrate meaningful or statistically reliable associations with elevated depression. The only other notable predictors are Hip hop and Music effects: Worsen, which show significantly higher odds of high depression compared to their reference groups.

Limitations

Our analysis has several important limitations that should be kept in mind when interpreting the results.

- **Self-reported and cross-sectional data:** All variables in the dataset, including mental-health symptoms and daily listening habits, were self-reported. This introduces potential measurement error due to inaccurate recall, inconsistent interpretation of questions, or social desirability bias. Additionally, because the data is cross-sectional, we cannot determine temporal ordering or causality. For instance, even if hours of listening were associated with depression, we cannot know whether listening more contributes to symptoms or whether individuals with higher depression simply listen to more music.
- **Limited measurement of key variables:** Several important constructs were captured using single questions (for example, music effects, listening to music while working, and favorite genre), which restricts measurement precision. Some mental-health scores may also overlap conceptually, as illnesses like anxiety and insomnia strongly correlate with depression. This makes it difficult to isolate unique events even when multicollinearity diagnostics appear acceptable.
- **Simplified representation of music behavior:** Daily hours of listening is an extremely limited measure of music engagement. It does not capture when people listen (morning vs. late night), how they listen (active vs. passive), why they listen (mood regulation, entertainment, background noise), or what they listen to within genres, all of which could meaningfully relate to mental-health outcomes.

Similarly, “primary streaming service” does not capture multi-platform use or variability in content algorithms.

- **Missing potential confounders:** Several important correlated variables of depression, like socioeconomic status, employment, physical health, sleep patterns, chronic stress, and academic workload, were not measured. Their absence means our model may suffer from omitted-variable bias, limiting the interpretability of estimated relationships.

Future Work

To strengthen the results and analysis, several steps could be taken in future research:

- **Collect richer and more detailed data:** Future studies would benefit from using validated multi-item psychological scales, such as PHQ-9 for depression or GAD-7 for anxiety, rather than the single self-reported questions on a scale of 0-10. Incorporating more granular measures of music engagement, such as listening context, emotional motivation, time-of-day patterns, or track-level characteristics, would also allow for more accurate modeling of how different aspects of music behavior relate to mental-health outcomes.
- **Use longitudinal or experimental study designs:** Because the current dataset is cross-sectional, it cannot speak to temporal ordering or causal relationships. Longitudinal surveys could help determine whether music-listening habits change before shifts in mood or vice-versa. Experimental designs, such as one that involves assigning participants to specific listening conditions, would further clarify causal pathways.
- **Apply more flexible modeling approaches:** Although poisson and logistic regression provided interpretable results, more flexible models that are outside the scope of this class, like generalized additive models, random forests, or gradient-boosted trees, could capture non-linear relationships and interactions that traditional parametric models may miss.
- **Address unmeasured confounders and broaden co-variates:** Adding variables related to sleep quality, stress, socioeconomic status, work or academic pressure, or physical health would reduce omitted-variable bias. These variables are often meaningful predictors of depression and may help clarify whether music-listening habits have independent effects.
- **Expand the scope of musical data:** Streaming-platform logs or acoustic features from tracks (tempo, energy, valence, etc.) would enable more detailed analysis of how specific music properties or listening patterns relate to mental-health symptoms. This would move the analysis beyond total hours per day and allow for deeper behavioral insights.

These limitations mean that our findings should be interpreted as descriptive patterns in this sample, rather than definitive evidence about the causal effect of music listening on depression. Nonetheless, the analysis provides a useful starting point for understanding how music-listening habits and mental-health symptoms co-vary in this population.

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